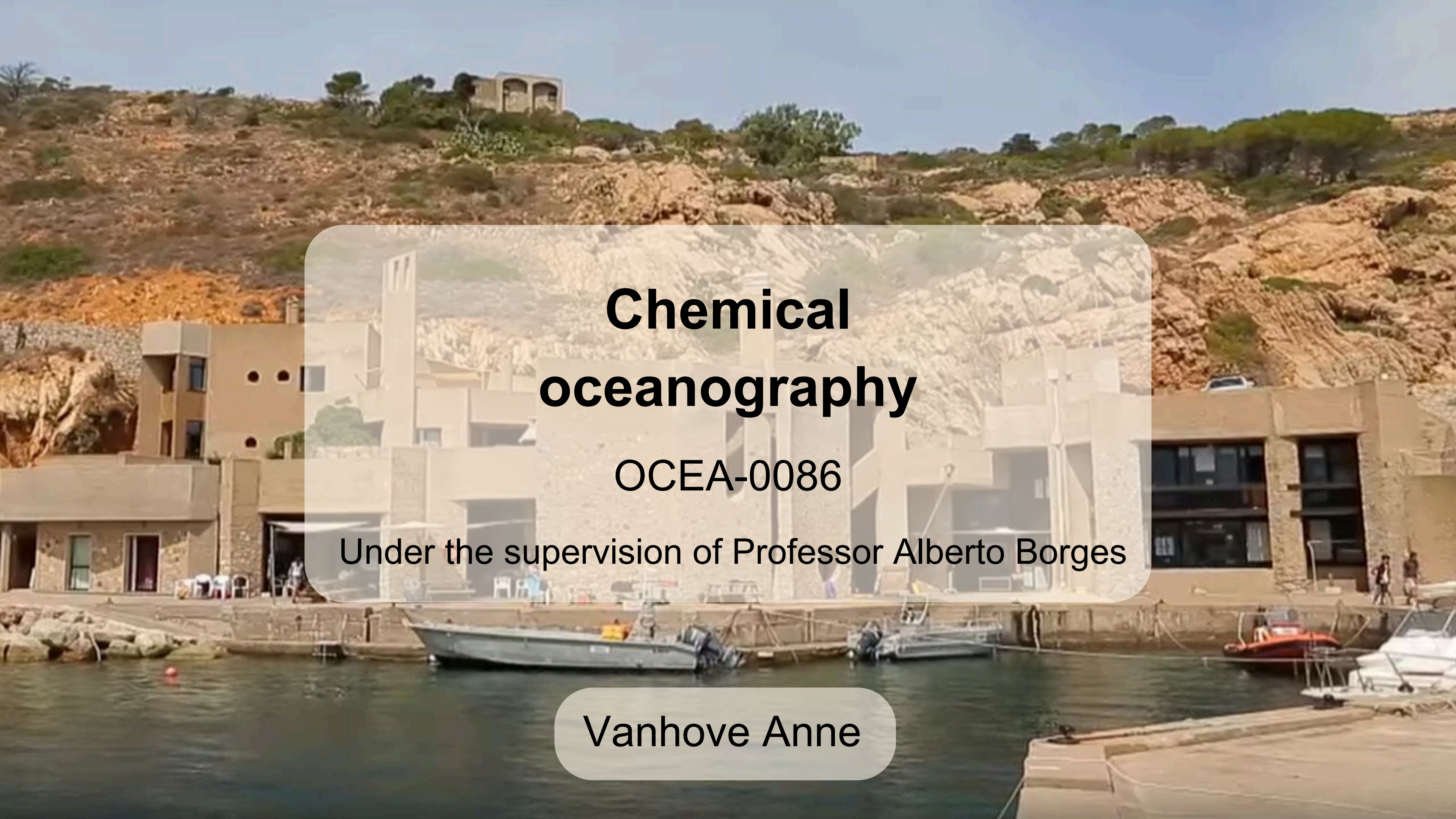


The background image shows a coastal town with a hillside in the background. The hillside is covered in sparse vegetation and has a small building on top. In the foreground, there is a harbor with several boats docked. The water is calm, and the sky is clear. The town's buildings are built into the hillside, and there are some people visible on the waterfront.

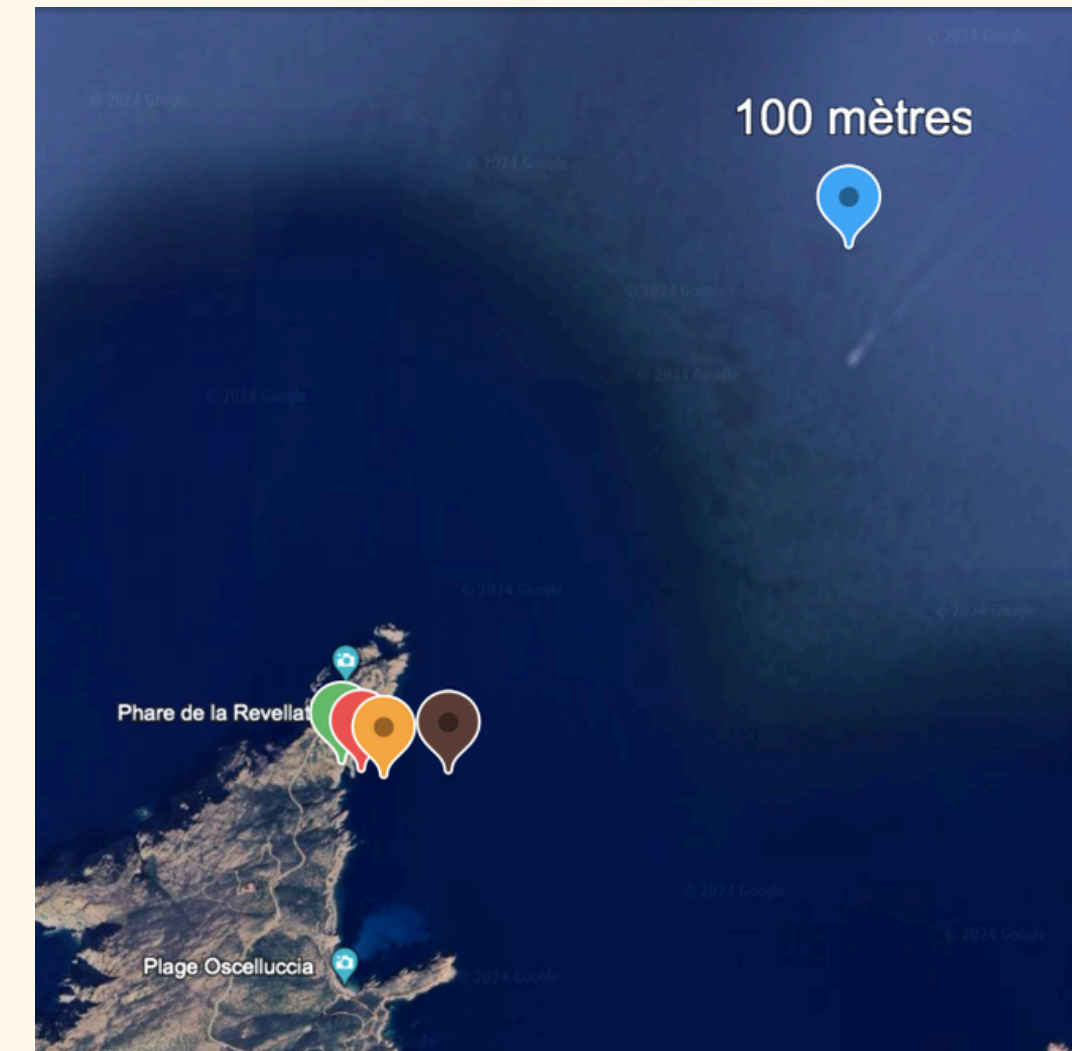
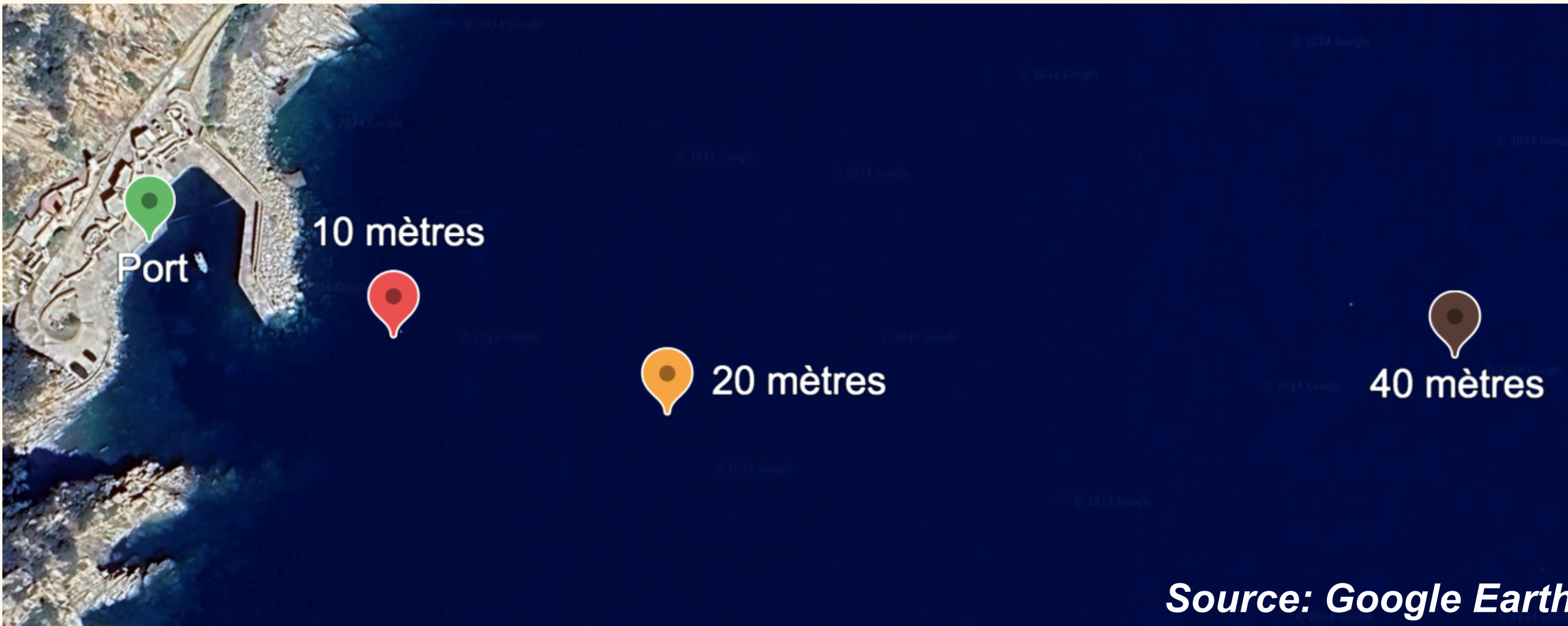
Chemical oceanography

OCEA-0086

Under the supervision of Professor Alberto Borges

Vanhove Anne

Methodology :



5 stations:

- Port: 1m et 2m
- 10 meters: 1m, 5m et 10m
- 20m, 40m et 100m: 1m then every 10 meters

For each station:

CTD + Niskin

Temperature measurement:

- Temperature ↘ as depth ↗
- Presence of a slight thermocline within the first 5 meters --> surface waters
- 20–30 m: seasonal thermocline

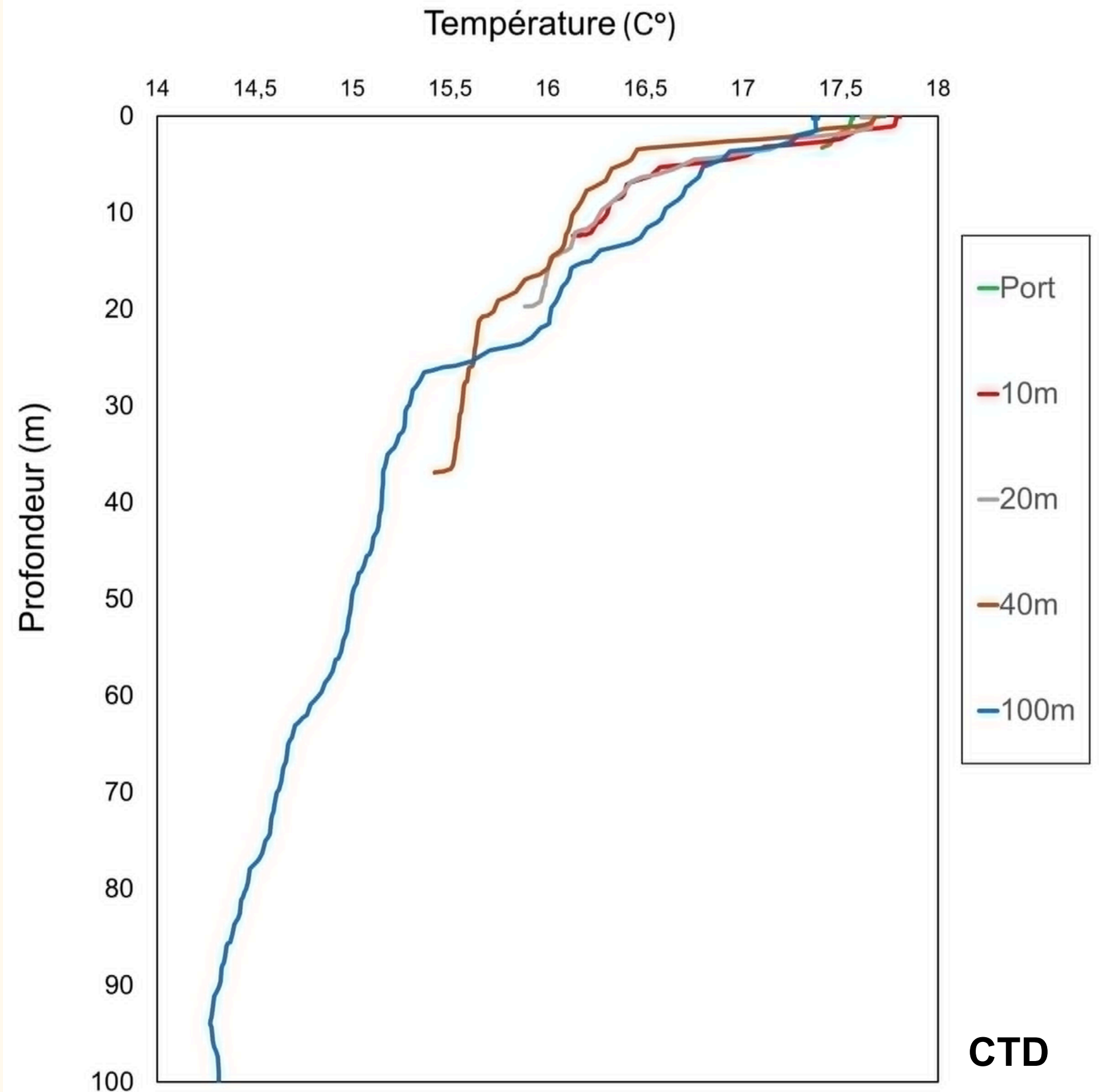
CTD:

- Highest temperature: 17.8°C
- Lowest temperature: 14.3°C
- > a drop of 3.5°C over 100 meters

Niskin

- T°max: 19,8°C T°min: 18,3°C
- > Not representative

--> temperature difference because the Niskin sample was measured later in the laboratory rather than in situ

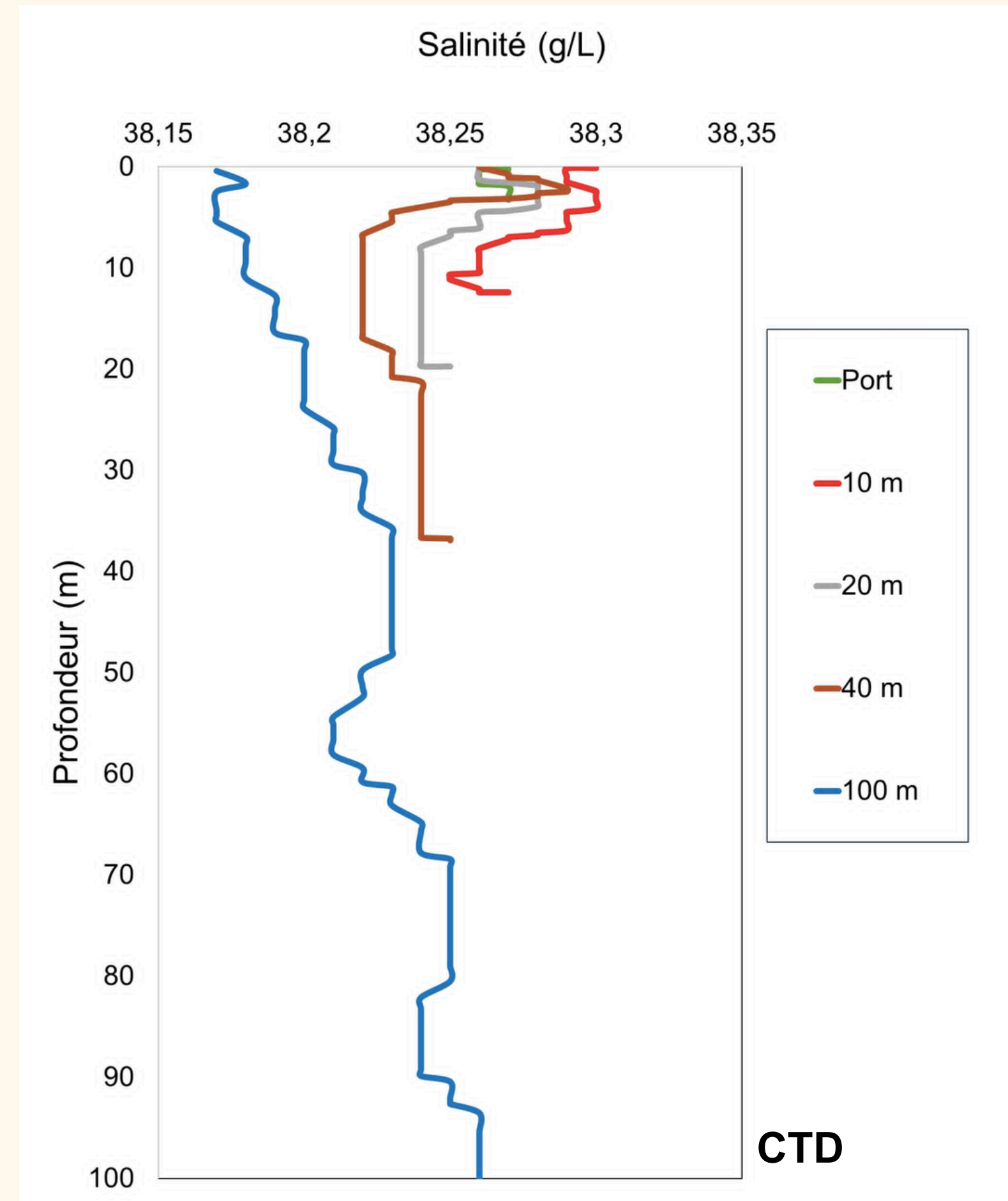


Measuring salinity:

- Port, 10 m, 20 m, and 40 m: salinity identical to that at the surface, with similar profiles at depth
- Lower salinity due to evaporation along the coast + more turbulent waters in the canyon

TDC:

- Typical shape of the normal curve, ↗ salinity with depth --> colder and saltier water = denser water



pH measurement:

Niskin bottle

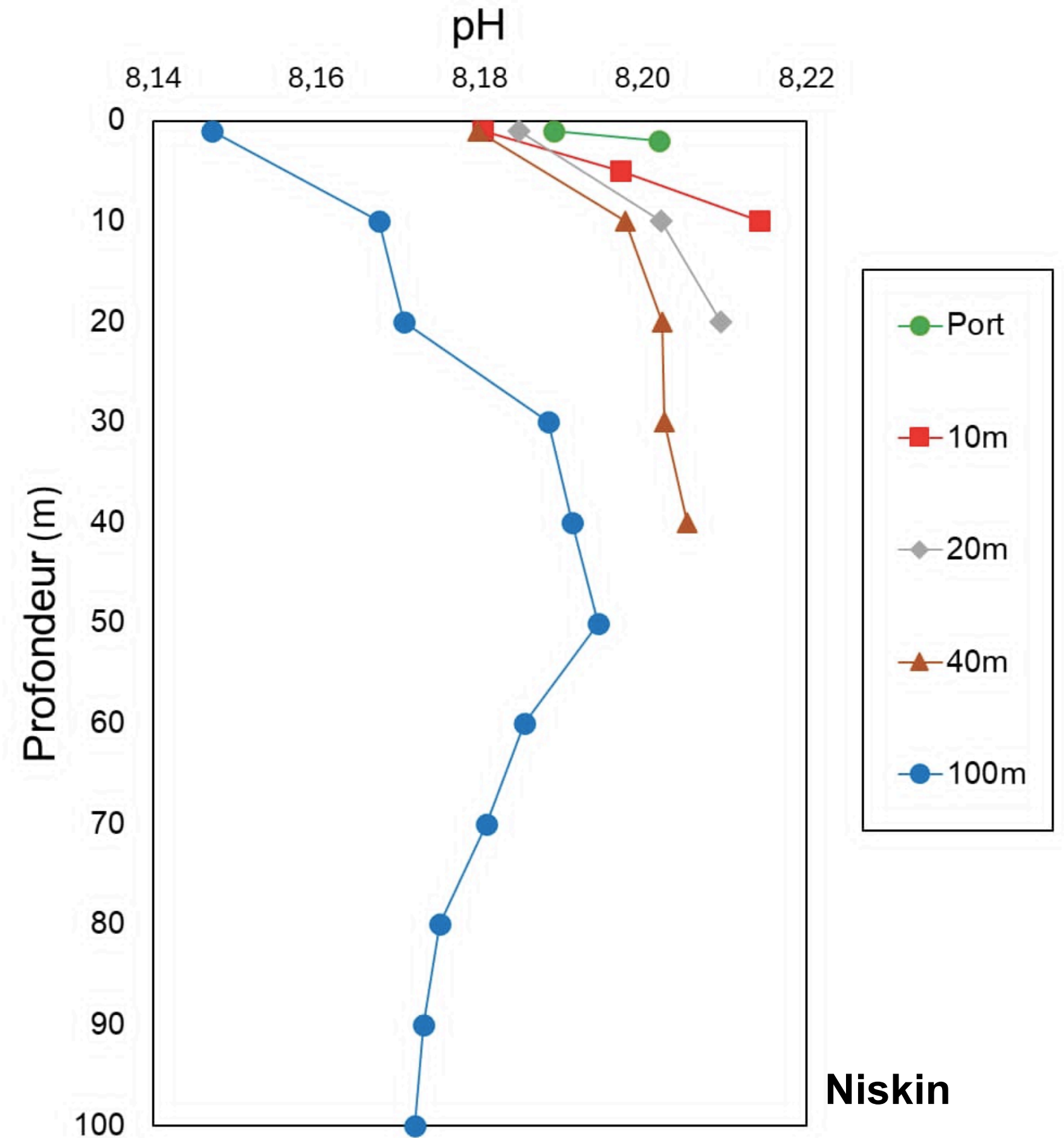
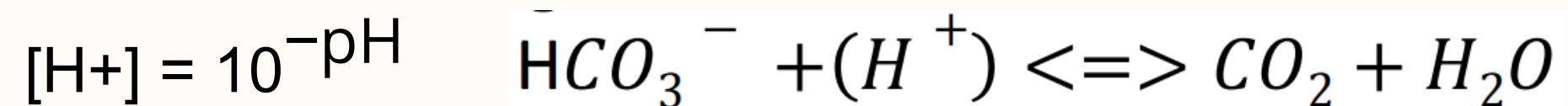
At the site level: pH ↗ toward the coast because Posidonia + algae + phytoplankton consume CO₂ --> ↘ H⁺ ions = more basic water

TDC: 0 to 50 meters: autotrophic organisms that use CO₂ for photosynthesis, resulting in less H⁺ in the environment

50 to 100 meters: respiration --> O₂ consumption + CO₂ production ==> more H⁺ = more acidic water

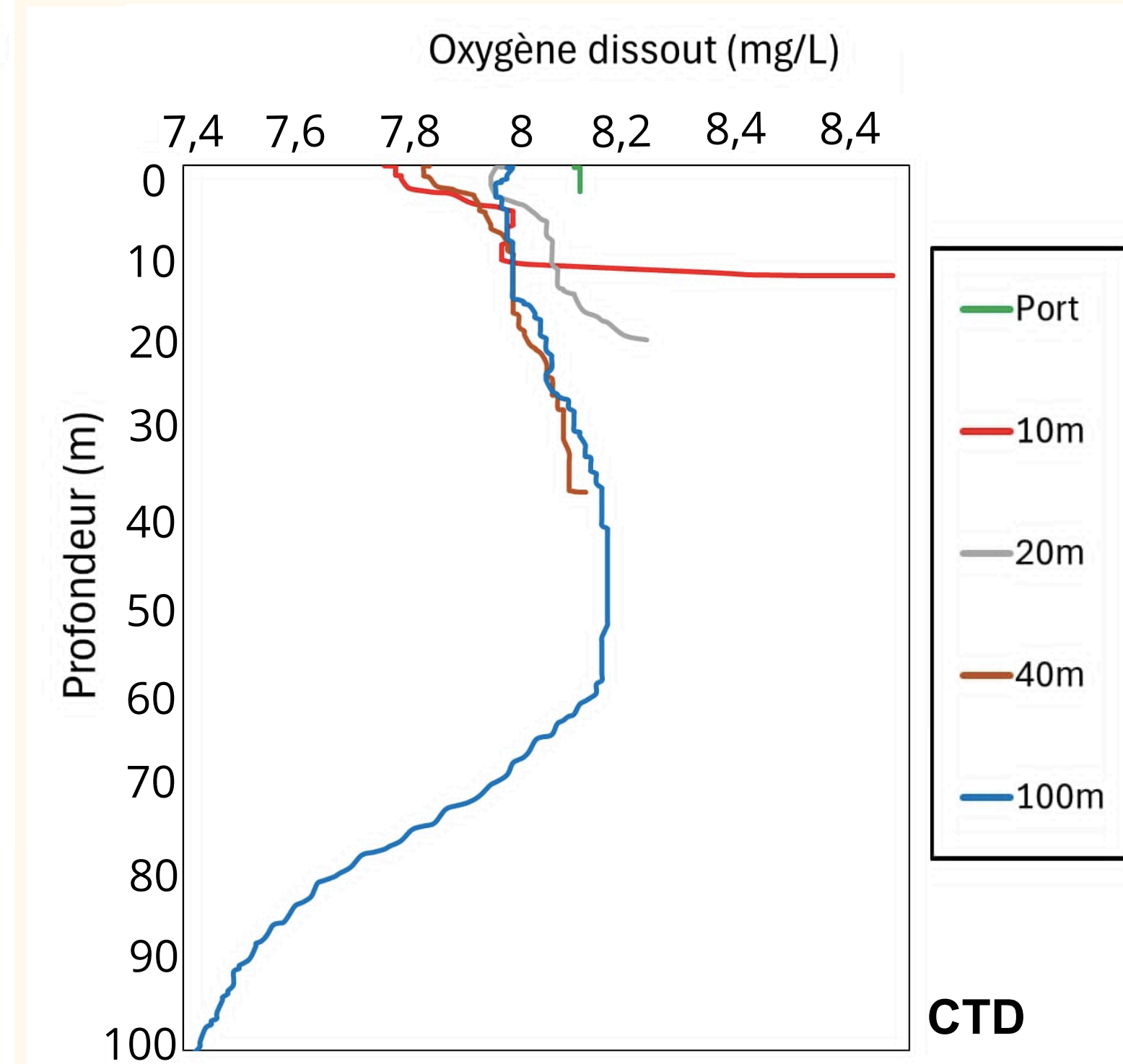
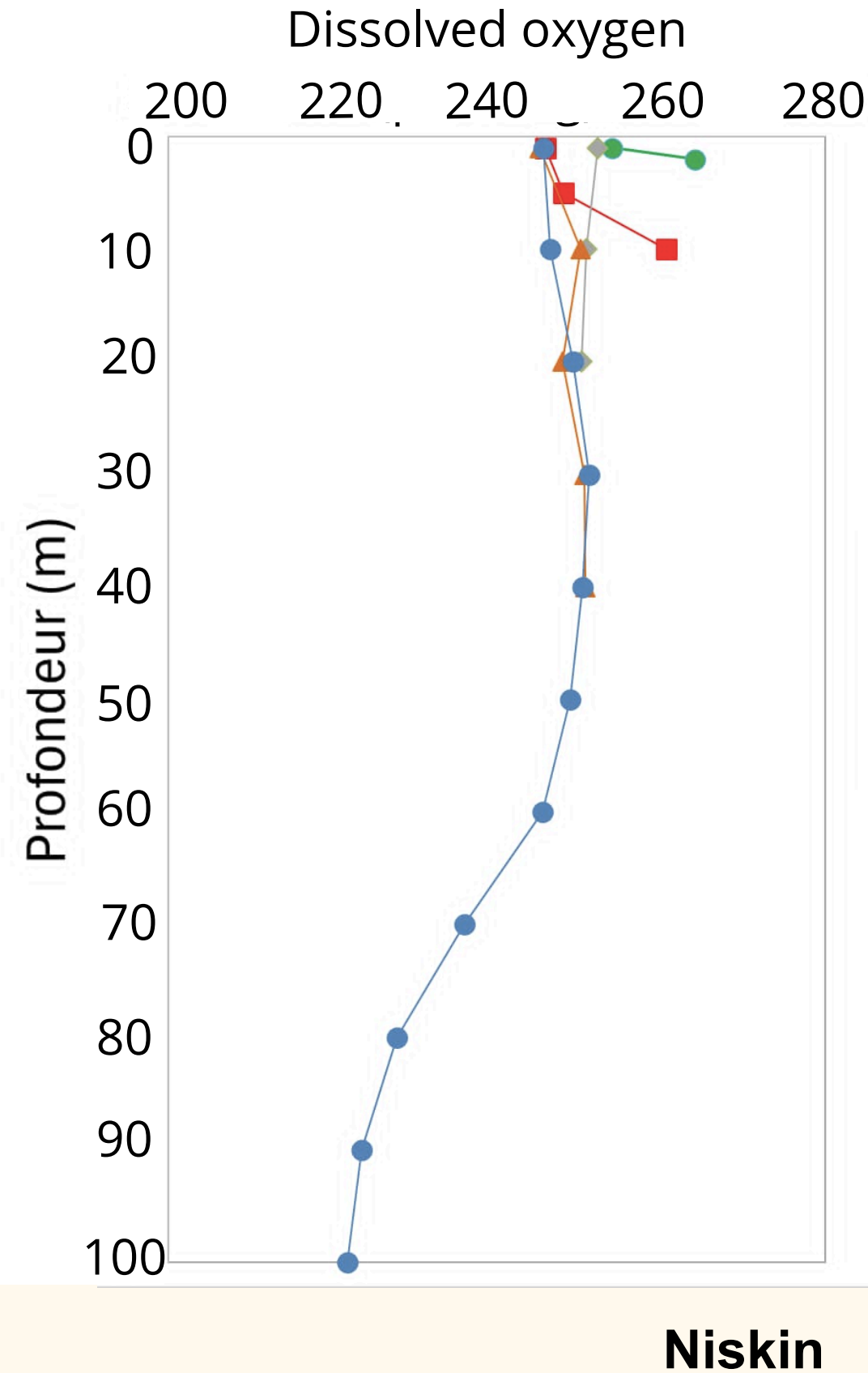
50 meters and deeper: consumption of organic matter

CTD: Unusable data, negative pH reading



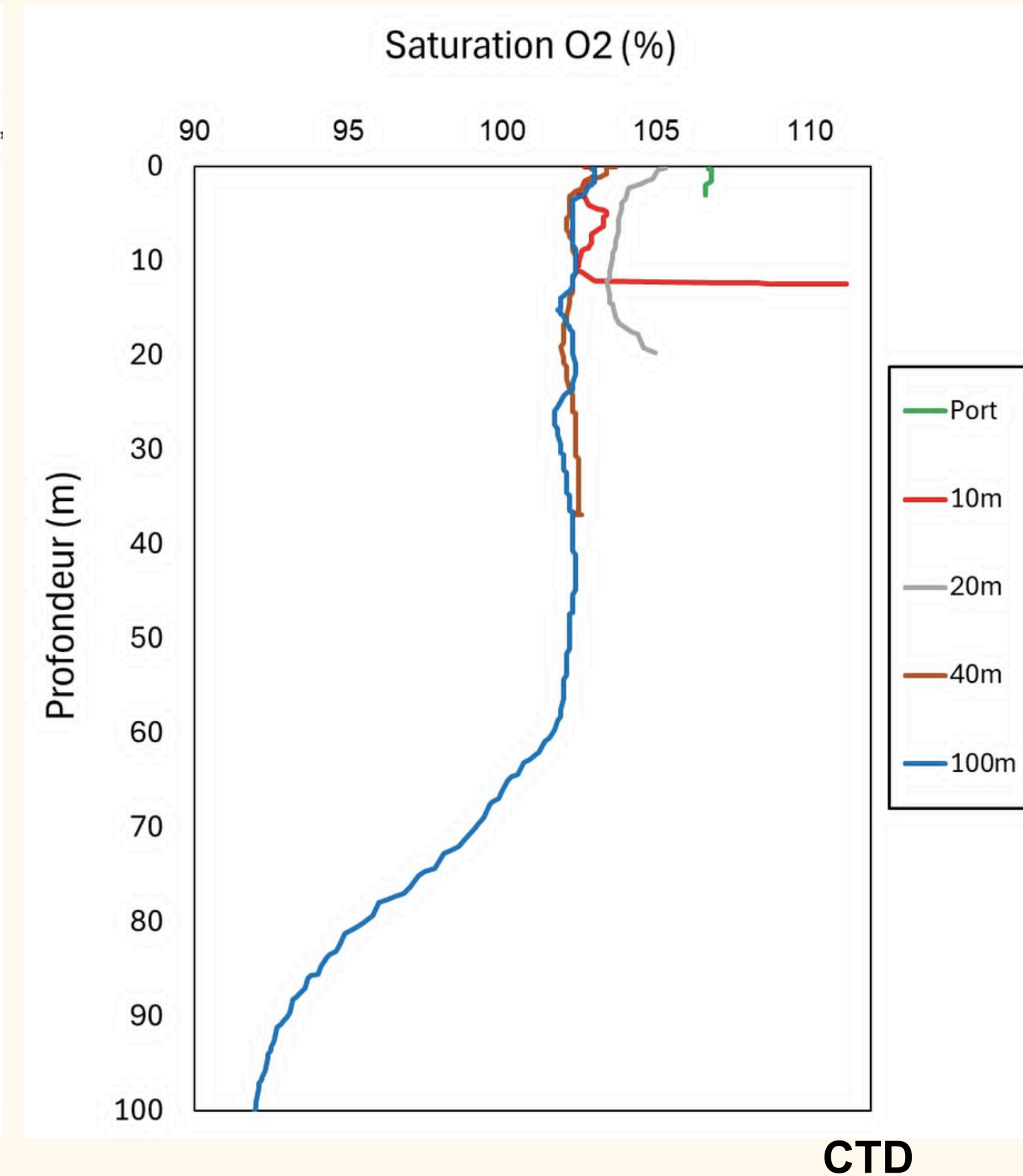
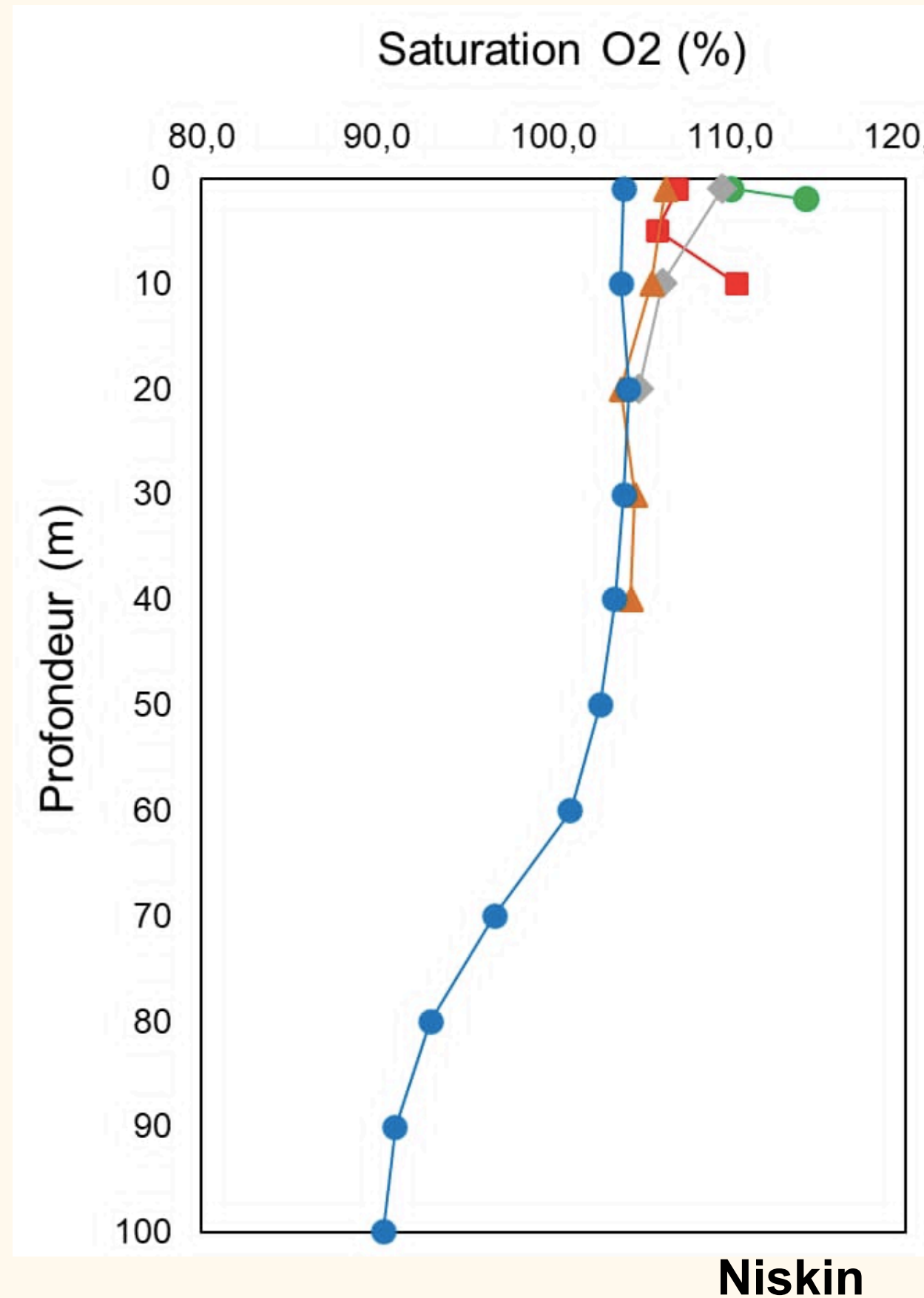
Measurement of dissolved oxygen concentration:

- Similar trends: ↗ steeply upward, then ↘ after 50 meters
- **After 50 m:** ↘
photosynthesis = ↘ O_2 production
- High levels of dissolved oxygen at the port due to the presence of Posidonia



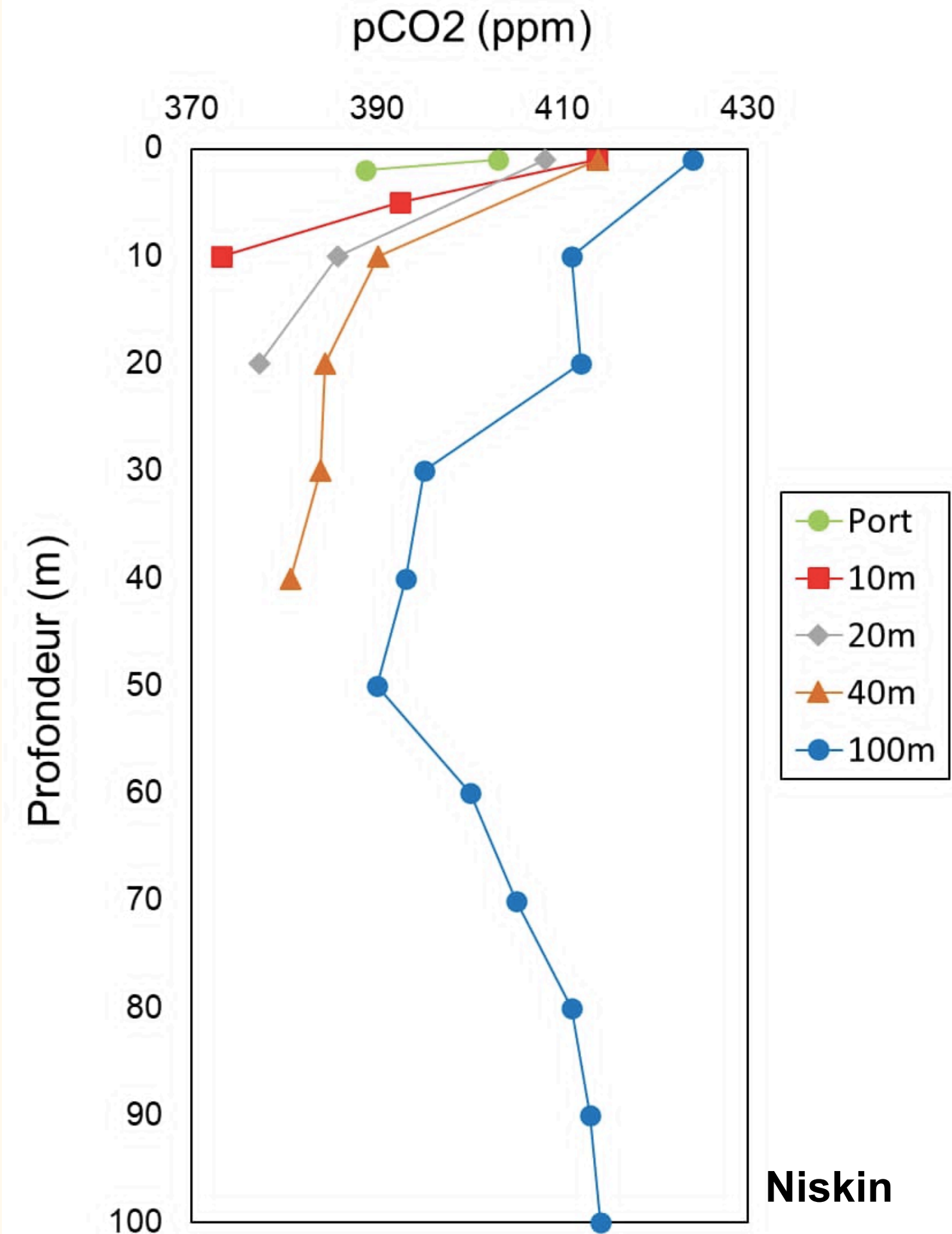
Measurement of saturated oxygen concentration:

- Saturation --> oxygen production or consumption
- ↗ coastal production = prod
- **0 to 60 m:** >100% = Oxygen production --> autotrophic organisms
- **60 to 100 m:** <100% --> breathing



Partial pressure of CO₂

- **Concentration decreases** as one approaches the coast --> photosynthesis by Posidonia
- **Surface - 40 m:** Photosynthesis by Posidonia, algae, and phytoplankton, which consume CO₂ and produce O₂ --> low [CO₂]
- **After 50 m:** organisms that breathe consume O₂ and produce CO₂ --> pH curve = acidification



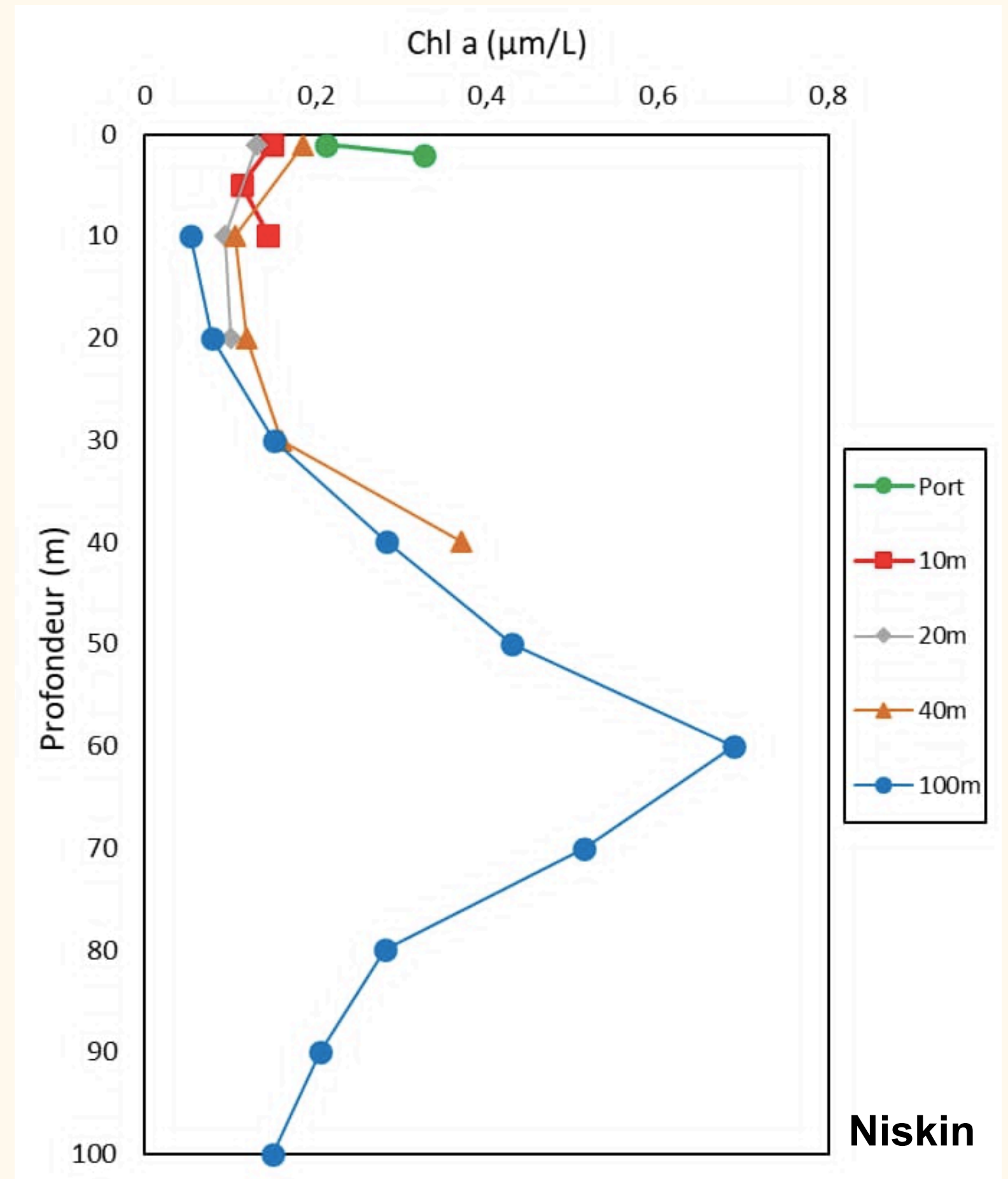
Measurement of chlorophyll a:

- **0 to 50 m:** maximum concentration of phytoplankton and algae due to good light conditions + radiation too weak to damage cells --> high chlorophyll a
- **From 50 to 100 m:** Chl decreases due to a decline in photosynthetic organisms and an increase in light energy absorption
- **Canyon head:** measured data from 0 to 10 meters are unusable because they are negative

Correlation:

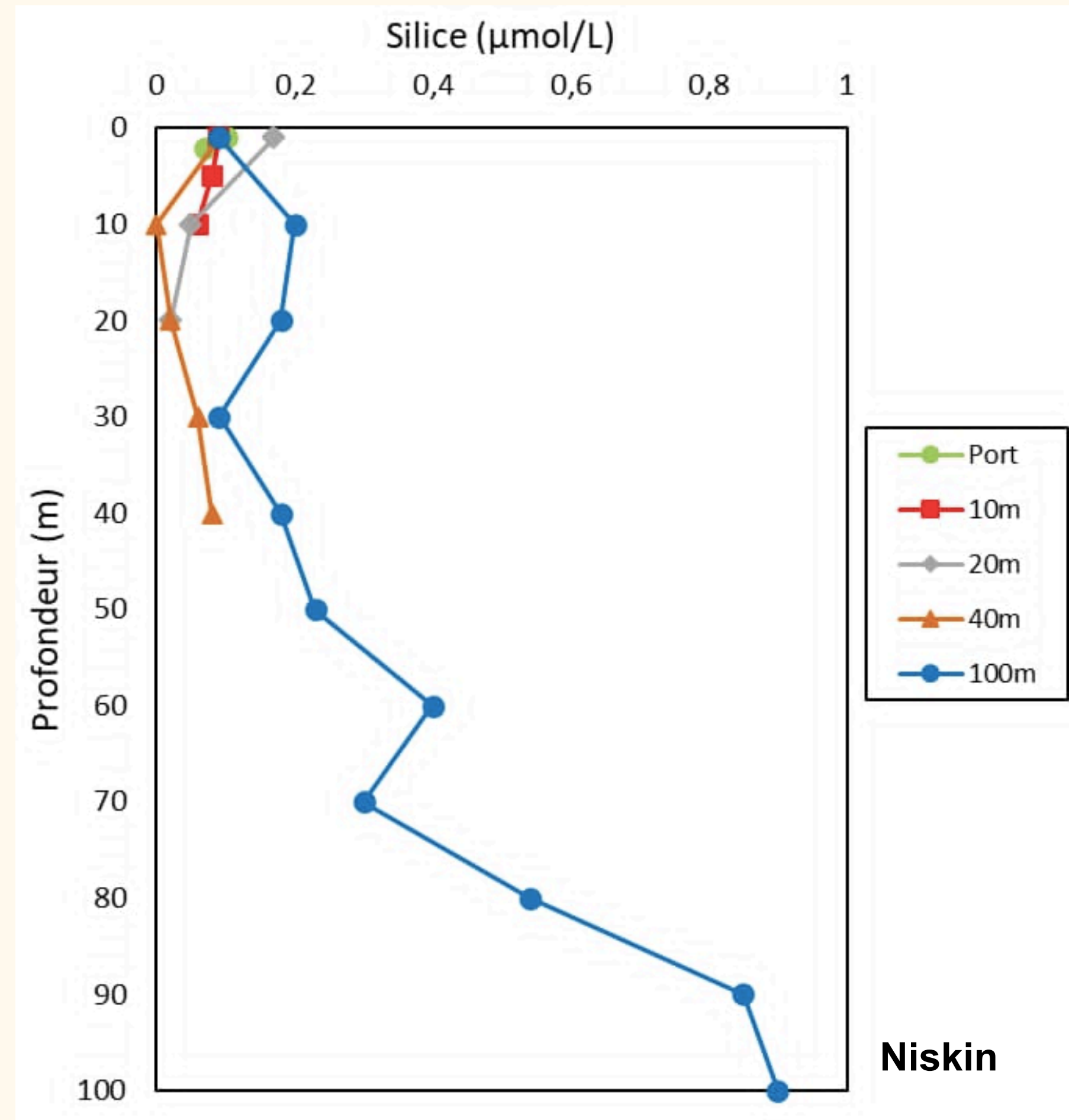
Surface: Chlorophyll a, dissolved O_2 , pCO_2 --> photosynthesis = more alkaline pH

Depth: ↘ photosynthesis, ↘ dissolved O_2 , ↗ respiration, ↗ dissolved CO_2 + nitrification = H^+ production = more acidic pH



Silica

- A major component of certain organisms
- **Surface:** ↘ [silica] because it is consumed by siliceous organisms (diatoms)
- **30–100 m:** sedimentation of silicate particles and organisms --> dissolution of silica --> ↗ [silica]

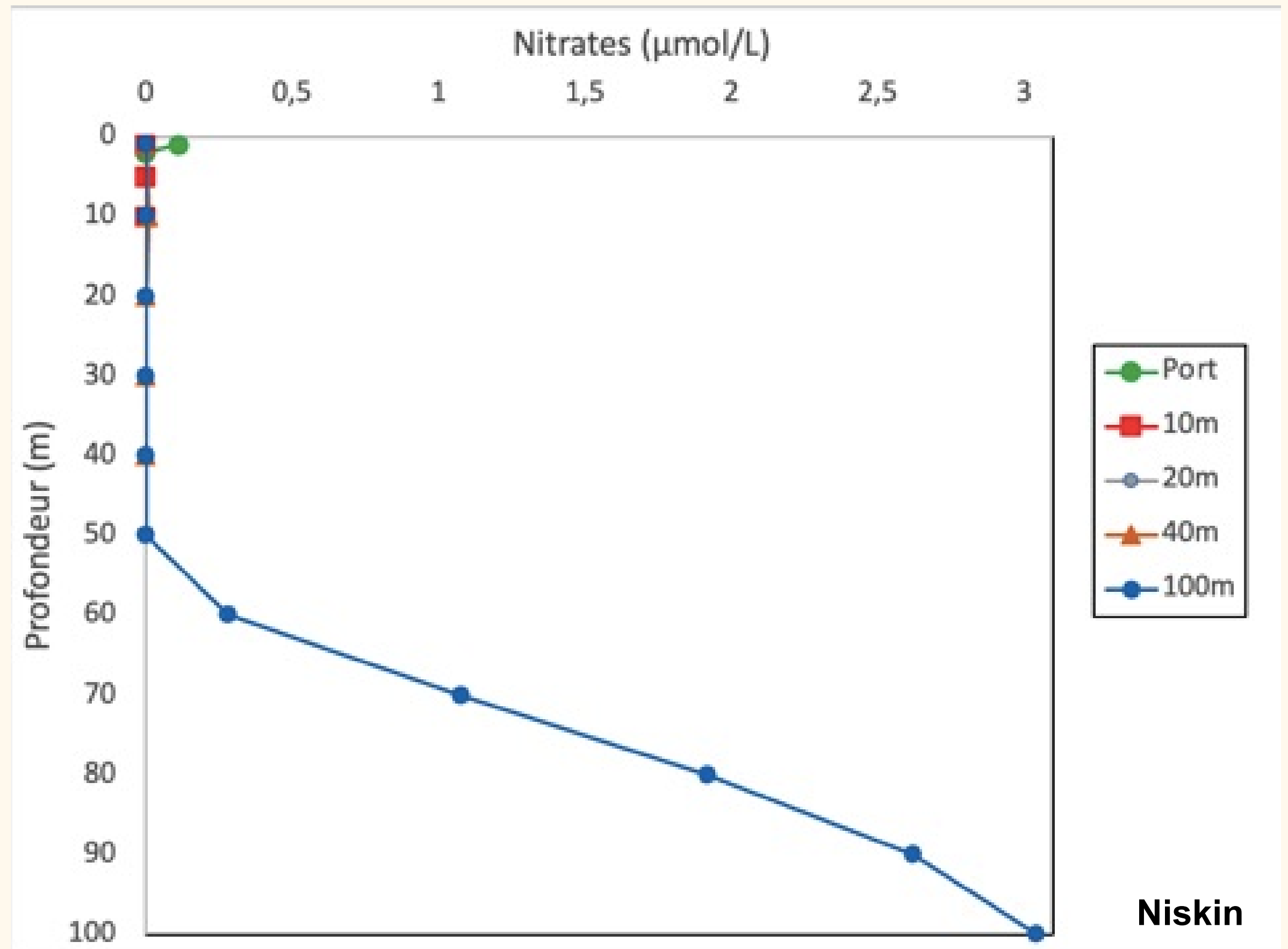


Nitrites and nitrates

- Nitrate and nitrite curves with a similar pattern
- **0–100 m:** Value too low to be detected --> oligotrophic sea, nitrate depletion (bloom)
- **50–100 m:** less phytoplankton, so fewer nutrients are consumed --> increased levels of nitrates and nitrites = nitracline

MO decomposition = production of ammonium, nitrites, and nitrates

--> follows the pH curve because H^+ production = acidification



Other nutrients

Nitrite and nitrate: similar

Ammonium: zero readings -->
measurement error, blank error, or
value too low to be detected
--> **oligotrophic** sea

Phosphate: measurement error at
the 40-meter station
More abundant at the surface due
to terrestrial inputs
Limited by phosphorus because the
FePO₄ complex (iron
orthophosphate) forms particles
(salts) --> these settle and remove
phosphorus from surface waters -->
maximum values at 80, 90, and 100
meters

station	Depth (m)	Nitrate (μmol/L)	Nitrite (μmol/L)	Ammonium (μmol/L)	Phosphate (μmol/L)
100m	1	0	0,02	0	0,05
	10	0	0,03	0	0,01
	20	0	0,03	0	0
	30	0	0,03	0	0,01
	40	0	0,04	0	0
	50	0	0,03	0	0
	60	0,28	0,08	0	0,03
	70	1,08	0,19	0	0
	80	1,92	0,27	0	0,02
	90	2,62	0,26	0	0,05
	100	3,04	0,18	1,42	0,07
40m	1	0	0,02	0	0,07
	10	0,01	0,01	0	0
	20	0	0,02	0	0
	30	0	0,02	0,22	4,87
	40	0	0,03	0	0
20m	1	0	0	0	0,06
	10	0	0	0	0
	20	0	0	0	0
10m	1	0	0	0	0,01
	5	0	0,01	0	0,03
	10	0	0	0	0,03
Port	1	0,11	0,01	0	0
	2	0	0,01	0	0,01